Package 'BOLDx'

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Type Package

Title Dose and Intervention Planning with Trial Simulation

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Description Software for dose and intervention planning and trial simulation in Bayesian ordered lattice dose finding, with prediction-guided prevention, delayed toxicity outcomes, or both. Includes complete-follow-up BOLD and BOIN comparators, a time-to-event BOIN comparator, configurable Beta priors, sequential child-specific mixture weights, observed-history cohort dose recommendations, intervention planning flags, and Monte Carlo operating characteristics. Extends the design of Wang and Tatsuoka (2026) <doi:10.1002/sim.70456>.

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`beta_mixture` *Inspect sequential weighted Beta components*

Description

Compute mixture weights, Beta component parameters, the mixture mean, and the overdose probability for prediction and partial-follow-up histories.

Usage

```
beta_mixture(
  a,
  b,
  q = numeric(),
  labels = rep(-1L, length(q)),
  sensitivity = 0.9,
  specificity = 0.98,
  target = 0.25
)
```

Arguments

a, b	Positive Beta parameters after resolved, nonlatent outcomes.
q	Remaining DLT-window fraction for each latent patient in [0,1]. Use 1 for an intervention patient without an observed DLT.
labels	Prediction labels: 1 positive, 0 negative, -1 uninformative.
sensitivity, specificity	Prediction operating characteristics.
target	Tail threshold for computing overdose probability.

Details

Components are weighted by sequential child-specific plug-in scores. Patient order is therefore part of the calculation. Known outcomes should first be incorporated into a and b. An observed intervention failure is a known DLT: increment a rather than branching.

Value

List containing components, mean, and overdose_probability.

Examples

```
beta_mixture(.75, 2.25, q = c(1, .5), labels = c(1L, -1L))
```

`bold_config` *Configure BOLD simulations and patient-history recommendations*

Description

The prior at dose j is Beta(prior_mean[j] * prior_pess[j], (1 - prior_mean[j]) * prior_pess[j]). True untreated risks are separate simulation inputs and are never supplied to dose-assignment calculations.

Usage

```
bold_config(
  target = 0.25,
  nmax = 21L,
  tau = 0.5,
  overdose_cutoff = 0.9,
  early_n = 12L,
  sensitivity = 0.9,
  specificity = 0.98,
  eligibility = 1,
  rho = 0.8,
  window = 3,
  accrual_rate = 2,
  arrival = c("uniform", "exponential"),
  timing = c("weibull", "uniform", "late"),
  prior_mean = NULL,
  prior_pess = 3,
  start_dose = 1L,
  pending_limit = 0.5,
  min_two = FALSE,
  gamma = overdose_cutoff
)
```

Arguments

`target`	Target untreated DLT probability, strictly between 0 and 5/7.
`nmax`	Patient cap, a positive multiple of three.
`tau`	BOLD decision-tail target in (0,1); not the DLT target.
`overdose_cutoff`	Posterior overdose elimination cutoff in (0,1).
`early_n`	Stop when this many patients have received the current dose and the next recommendation is unchanged. Use Inf to disable.
`sensitivity, specificity`	Prediction sensitivity and specificity for latent untreated DLT status, each in [0,1].
`eligibility`	Probability a patient is eligible for prediction.
`rho`	Probability an intervention prevents an otherwise-occurring DLT.
`window`	DLT assessment window, in months.

accrual_rate	Mean number of arrivals per month.
arrival	Arrival-gap distribution, uniform or exponential.
timing	Generative conditional DLT-time distribution. The working follow-up model remains uniform for all choices.
prior_mean, prior_pess	Dose-specific vectors, or scalars recycled to all doses. NULL prior_mean uses target; PESS defaults to 3.
start_dose	Starting dose index, validated against each scenario.
pending_limit	Maximum pending fraction at the current dose before suspension; defaults to 0.5.
min_two	Require two completed outcomes before escalation.
gamma	Alias for overdose_cutoff, default 0.90. Eliminate a dose when its posterior overdose probability exceeds this cutoff. If both names are supplied they must agree. Applies to simulations and recommendations.

Value

A validated configuration list of class bold_config.

Examples

```
cfg <- bold_config(nmax = 21, rho = 0.8)
cfg$target
```

BOLDx_workflows *Two-part guide: patient recommendations and simulation*

Description

BOLDx has two independent workflows: recommend a dose (and intervention when applicable) from observed patient records, or simulate operating characteristics under assumed true DLT risks. True risks are never inputs to observed-history recommendations.

Part 1

dose and intervention recommendations: Use plan_bold for a combined next-dose and intervention plan. Supply history, one row per enrolled patient in enrollment order, with dose, dlt and elapsed followup in months. Optional historical columns are prediction, eligible and intervened. Separate prediction and eligible arguments describe the incoming cohort of three. Supply ndoses, current_dose, config, and previously excluded doses. Use history=NULL to start. Original BOLD (bold), BOLD-prediction (prediction), TITE-BOLD (tite) and combined prediction/TITE (combined) all recommend doses. Only the two prediction-assisted designs flag preventive interventions.

The dose recommendation includes the action, next dose and reason, eliminated doses and dose-level summaries. Actions are start, escalate, stay, deescalate, wait or stop. intervention_plan contains incoming-patient flags. ready is TRUE only if a dose is assigned and required incoming predictions are available. A wait/stop never schedules an intervention. Record actual interventions and outcomes in history before calling again; preserve excluded_doses. This is next-cohort allocation, not final MTD declaration.

For history with three patients at dose 1, no observed DLTs and three months of follow-up each, default settings and incoming predictions (1,0,0) in prediction mode return escalate to dose 2, intervention flags (TRUE,FALSE,FALSE), and ready=TRUE. Without intervention, observed outcomes replace pending predictions. After intervention, no observed DLT leaves untreated status latent; an observed failure is a known DLT. Combined mode waits for intervention-patient resolution. External classifiers are not fitted by this package. Protocol-specific review is required before clinical use.

Part 2A

one specified risk scenario: Use `simulate_bold` with p.true (or truth), a nondecreasing numeric vector of true untreated DLT probabilities. For example, p.true=c(0.05,0.15,0.30,0.45,0.60), target=0.30 and mtd=3 define one scenario. The vector is user-specified, not randomly generated. The default nsim=1000 replicates trials for each requested method.

Outputs are summary (point estimates), mcse (within-scenario Monte Carlo SE) and trials (individual simulated records). Metrics are accuracy_pct, mean_n, actual_dlts, above_mtd_pct and duration_months. The last is displayed for TITE, combined and TITE-BOIN, and is NA for complete-follow-up designs. Actual DLTs are observed post-intervention events. Above-MTD allocation is the patient-weighted percentage, not an unweighted mean of trial ratios.

Reproducible example: with target=.30, cap 21, gamma=.90, tau=.50, sensitivity=.90, specificity=.98, rho=.80, seed=42 and 1000 trials, Original BOLD returns accuracy 51.20 percent (MCSE 1.58 percentage points), mean N 20.598 (MCSE 0.040) and actual DLTs 6.000 (MCSE 0.046). These illustrate this setting only, not universal comparative performance.

Part 2B

random scenarios from gaps: Use `simulate_random_scenarios`. Defaults are ndoses=4, mtd=2, lower_gap=.10, upper_gap=.15, risk_bounds=c(.01,.70), nscenarios=50 and nsim=1000. For each interior MTD j, the generator fixes its risk at target and its neighboring risks at target-lower_gap and target+upper_gap. Remaining risks are sorted uniform draws in the tails. Thus the default curves are (.15,.25,.40,U), with U between .40 and .70. See `generate_scenarios` for boundary-category rules.

Outputs include scenarios, settings, per-scenario summary and mcse, individual trials, and aggregate. Aggregate reports equally weighted mean and SD of scenario-level estimates, separately by method, MTD and gaps. With mtd=2, upper_gap=.15, default config, 50 scenarios, 1000 trials and seed=42, Original BOLD has mean accuracy 50.02 percent (scenario SD 1.61 percentage points), mean N 19.43 (SD .11) and actual DLTs 5.24 (SD .08).

Vector MTD categories and gaps evaluate all combinations. For mtd=0:4 and upper_gap=c(.15,.20), 50 scenarios per setting and 1000 trials mean 500000 trials per method. MTD 0 is all-too-toxic; ndoses+1 is all below target. A boundary gap without a neighbor has no effect. Some settings have no free doses and yield repeated curves. This uniform-tail generator is explicitly defined; it does not recreate saved manuscript scenarios. Use `generate_scenarios` to generate curves without trials.

Part 2C

your own scenario matrix: Use `simulate_scenarios`, one risk vector per matrix row. Supply explicit MTD labels for all-too-toxic or below-target cases. Otherwise omitted mtd selects the dose closest to target (lower dose breaks ties). Evaluation labels never enter allocation decisions.

Distinguish SE from SD

Within one scenario, mcse describes Monte Carlo uncertainty from the simulated trials. For means of N, DLTs or duration, SE is trial-level SD divided by sqrt(nsim). Accuracy SE uses the correct-selection indicator times 100. Above-MTD SE uses ratio influence values. Raw trials allow custom trial-level SD calculation.

Across a panel, aggregate SD describes dispersion among scenario-level estimates and includes finite-trial noise. It is not an SE. For 50 scenarios with 1000 replicates, this SD uses the 50 estimates, not the 50000 individual trial records. One-scenario groups have undefined SD.

Shared settings

bold_config sets target=.25, gamma=.90 and tau=.50 by default. Gamma controls posterior overdose elimination; tau controls the BOLD allocation-tail target. Both can be changed for either workflow. gamma is an alias for overdose_cutoff and is stored under that name. Other inputs include patient cap nmax, sensitivity, specificity, eligibility, rho, prior means/PESS, follow-up window, accrual, timing and pending rules. Simulation rho and eligibility generate histories; recommendations use the actual entered records, not simulated unknown outcomes.

See Also

plan_bold, simulate_bold, simulate_random_scenarios, bold_config

Examples

```
cfg <- bold_config(gamma=.90, tau=.50)
h <- data.frame(dose=c(1,1,1), dlt=c(0,0,0), followup=c(3,3,3))
p <- plan_bold(h, 4, 1, method="prediction", config=cfg,
               prediction=c(1L,0L,0L))
p$dose_recommendation
p$intervention_plan
## Not run:
methods <- c("bold","prediction","tite","combined")
single <- simulate_bold(p.true=c(.05,.15,.30,.45,.60),
  config=bold_config(target=.30), method=methods, nsim=1000, seed=42, mtd=3)
single$summary
single$mcse
panel <- simulate_random_scenarios(mtd=2, upper_gap=.15,
  config=cfg, method=methods, nscenarios=50, nsim=1000, seed=42)
panel$scenarios
panel$aggregate

## End(Not run)
```

---

generate_scenarios *Generate ordered toxicity scenarios from adjacent-dose gaps*

Description

Generate a reproducible panel for one MTD category and gap combination. This is a defined uniform-tail generator, not a claim to sample all possible dose-toxicity curves or to reproduce previously saved manuscript scenarios.

Usage

```
generate_scenarios(
  ndoses = 4L,
  target = 0.25,
  mtd = 2L,
  lower_gap = 0.1,
  upper_gap = 0.15,
  risk_bounds = c(0.01, 0.7),
  nscenarios = 50L,
  seed = 1L
)
```

Arguments

ndoses	Number of dose levels, 2 to 10; default 4.
target	Target untreated DLT probability; default 0.25.
mtd	MTD category. For 1:ndoses, fix that dose at target. Zero means all too toxic; ndoses+1 means all doses below target. Default 2.
lower_gap, upper_gap	Positive adjacent-dose risk gaps, default 0.10 and 0.15. For an interior MTD j, set p[j-1]=target-lower_gap and p[j+1]=target+upper_gap when those neighbors exist.
risk_bounds	Lower and upper limits for generated risks, default c(0.01,0.70). Must lie strictly inside (0,1) and contain target.
nscenarios	Number of scenarios for this category and gap setting, default 50.
seed	Nonnegative integer seed; the caller's random state is restored.

Details

Remaining lower doses are sorted uniform draws between the lower risk bound and the fixed lower neighbor; remaining upper doses are sorted uniform draws between the fixed upper neighbor and upper bound. For mtd=0, the first risk is target+upper_gap and remaining risks are uniform draws above it. For mtd=ndoses+1, the last risk is target-lower_gap and remaining risks are uniform draws below it. Gaps requiring a neighbor must leave a nonempty interval inside risk_bounds. Some boundary settings have no free doses and produce identical curves; inspect the matrix.

Value

Numeric matrix of true risk vectors, one scenario per row.

Examples

```
generate_scenarios(nscenarios=3, mtd=2, upper_gap=.15, seed=42)
```

plan_bold *Plan the next dose and prediction-guided intervention together*

Description

Combine an observed-history dose recommendation with intervention flags for incoming patients. Incoming predictions never alter the historical posterior.

Usage

```
plan_bold(
  history = NULL,
  ndoses,
  current_dose = NULL,
  method = c("bold", "prediction", "tite", "combined"),
  config = bold_config(),
  excluded = integer(),
  prediction = NULL,
  eligible = TRUE
)
```

Arguments

history, ndoses, current_dose, config, excluded	Arguments passed to recommend_dose().
method	One of bold, prediction, tite or combined.
prediction	External binary predictions for the incoming cohort, or NULL when predictions have not yet been obtained. A supplied cohort must contain three predictions; NA denotes an unavailable prediction.
eligible	Logical scalar or vector for the incoming cohort.

Details

Original BOLD and TITE-BOLD do not use preventive intervention. Prediction and combined modes flag eligible positive predictions. If predictions depend on dose, obtain them at the returned candidate dose before treating patients. This function neither trains a classifier nor supplies a final MTD declaration. Protocol-specific statistical review and validation are required before clinical application.

Value

A list with dose_recommendation, intervention_plan and ready. intervention_plan contains the candidate dose, intervene flags and status. No intervention is scheduled when dose allocation is waiting or stopped. ready is TRUE only when a dose is assigned and all required incoming prediction decisions are available. Actual interventions must subsequently be recorded in history, not inferred from these proposed flags.

Examples

```
cfg <- bold_config(gamma=.90, tau=.50)
h <- data.frame(dose=c(1,1,1), dlt=c(0,0,0), followup=c(3,3,3))
p <- plan_bold(h, 4, 1, method="prediction", config=cfg,
               prediction=c(1L,0L,0L))
p$dose_recommendation
p$intervention_plan
```

---

plan_intervention *Flag intervention eligibility from external predictions*

Description

Apply the modeled positive-prediction intervention policy. This function does not fit a classifier or perform an intervention.

Usage

```
plan_intervention(prediction, eligible = TRUE)
```

Arguments

prediction	Vector of binary predictions (0/1); NA means unavailable.
eligible	Logical vector, or scalar recycled to all predictions.

Value

A data frame containing prediction, eligible, intervene (logical, NA if an eligible patient's prediction is unavailable), and status.

Examples

```
plan_intervention(c(1L,0L,NA_integer_),eligible=c(TRUE,TRUE,FALSE))
```

---

recommend_dose *Evaluate the next cohort dose from entered patient histories*

Description

Apply the same dose-allocation rules used by the simulator to observed patient-level data. No true dose-toxicity probabilities are required.

Usage

```
recommend_dose(
  history = NULL,
  ndoses,
  current_dose = NULL,
  method = c("bold", "prediction", "tite", "combined", "boin", "tite_boin"),
  config = bold_config(),
  excluded = integer()
)
```

Arguments

history	Data frame in enrollment order with dose, dlt and followup columns. dlt=1 means a DLT has been observed; dlt=0 means none observed to date. NA is permitted only before follow-up reaches the assessment window. followup is elapsed follow-up in months. Optional prediction is 0, 1 or NA; eligible and intervened are logical columns. If omitted, eligible is inferred from prediction availability and intervened is FALSE. Use NULL for no enrolled patients.
ndoses	Number of ordered dose levels, from 2 to 10.
current_dose	Dose of the most recently enrolled cohort; required for nonempty history. For empty history, defaults to the configured start dose.
method	Design to evaluate: bold, prediction, tite, combined, boin, or tite_boin. Default is Original BOLD.
config	Design settings from bold_config().
excluded	Previously eliminated dose indices. Pass the previous result's excluded_doses to preserve irreversible elimination.

Details

Evaluation is supported at cohort boundaries of three patients. Original BOLD and prediction-assisted complete-follow-up BOLD wait until observed follow-up resolves. TITE methods use partial follow-up. Combined mode additionally waits for intervention patients to resolve.

Without intervention, a DLT or completed non-DLT follow-up replaces any pending prediction contribution. After intervention, an observed DLT is a known failure; no observed DLT leaves untreated status latent even after full follow-up. An intervention must have a positive prediction and an eligible patient. Actual intervention decisions must be explicitly recorded.

sensitivity and specificity weight available predictions. rho, eligibility probability, timing and accrual settings generate simulation histories; they do not replace observed patient records in this function. No unknown untreated outcomes should be entered. This interface implements the package's research protocol; application in a clinical trial requires protocol-specific statistical review and validation.

Value

A bold_recommendation list with action (start, escalate, stay, deescalate, wait or stop), next_dose (NA when waiting/stopped), reason, excluded_doses, dose_summary and patient_status. It is a next-cohort allocation result, not a final MTD selection.

Examples

```
h <- data.frame(dose=c(1,1,1), dlt=c(0,0,0), followup=c(3,3,3))
recommend_dose(h, ndoses=4, current_dose=1, method="bold")

hp <- transform(h, prediction=c(1L,0L,0L), eligible=TRUE,
                intervened=c(TRUE,FALSE,FALSE))
recommend_dose(hp, ndoses=4, current_dose=1, method="prediction")
```

`simulate_bold` *Simulate BOLD extensions and matched comparators*

Description

Evaluate dose-finding accuracy, enrollment, actual toxicity, allocation above the MTD, and calendar duration for BOLD extensions and their comparators.

Usage

```
simulate_bold(
  truth = NULL,
  method = c("prediction", "tite", "combined"),
  config = bold_config(),
  nsim = 1000L,
  seed = 1L,
  mtd = NULL,
  p.true = NULL
)
```

Arguments

`truth`	Nondecreasing vector of 2 to 10 untreated DLT probabilities, strictly between zero and one.
`method`	One or more of prediction, tite, combined, bold, boin, tite_boin.
`config`	A configuration from bold_config().
`nsim`	Number of replicate trials per method, default 1000.
`seed`	Nonnegative integer seed. The caller's random state is restored.
`mtd`	True optimal dose index for evaluation. NULL chooses the dose closest to target (lower index breaks ties). Set 0 for all-too-toxic; K+1 means all doses are below target and the highest dose is optimal. This value affects summaries only, never dose decisions.
`p.true`	Alias for truth, for a user-specified dose-toxicity scenario. Supply one name, or identical values under both names.

Details

Fixed cohorts of three are used. Intervention failure is an observed DLT, not an extra latent branch. In combined mode, intervention follow-up resolves before the next dose decision. Mixture weights use sequential child-specific plug-in scores; the full calculation and assumptions are documented in the package's online Methods document.

Value

A bold_simulation list containing summary, trials, config, truth, mtd, and seed. Summary contains scenario-level point estimates. Optional Monte Carlo precision diagnostics are retained separately in mcse.

Uncertainty

For one fixed scenario, mcse reports Monte Carlo standard errors across the nsim trials, not between-scenario SD. Accuracy SE is 100 times the sample SD of trial-level correct-selection indicators divided by sqrt(nsim). For mean N, actual DLTs and duration, SE is the trial-level sample SD divided by sqrt(nsim). Above-MTD SE uses the influence values of the patient-weighted ratio, not an unweighted mean of trial ratios. Raw trial records are retained in trials for other summaries.

Examples

```
z <- simulate_bold(c(.10, .25, .45, .60),
  method = c("prediction", "bold", "boin"), nsim = 10, seed = 1)
z$summary
```

---

simulate_random_scenarios

Simulate random scenario panels defined by toxicity gaps

Description

Generate and simulate nscenarios scenarios for each MTD and gap combination. Defaults are 50 scenarios and 1000 trials per scenario per method.

Usage

```
simulate_random_scenarios(
  ndoses = 4L,
  mtd = 2L,
  lower_gap = 0.1,
  upper_gap = 0.15,
  risk_bounds = c(0.01, 0.7),
  nscenarios = 50L,
  method = c("bold", "prediction", "tite", "combined"),
  config = bold_config(),
  nsim = 1000L,
  seed = 1L
)
```

Arguments

`ndoses`, `mtd`, `lower_gap`, `upper_gap`, `risk_bounds`, `nscenarios`
: Arguments to generate_scenarios(). Here mtd and either gap may be vectors; all distinct combinations are evaluated separately.

`method`, `config`, `nsim`, `seed`
: Arguments to simulate_bold(). The target probability is taken from config. Scenario seeds increment globally; a separate deterministic generation seed is used for each setting.

Details

Increasing nscenarios explores scenario heterogeneity; increasing nsim improves within-scenario Monte Carlo precision. Aggregate SD is not the SE of a mean and includes finite-trial simulation noise. Patient caps, predictor performance, rho and timing are set through config. To vary configurations, run this function separately and label results accordingly.

Value

A list with scenarios (matrix), settings (scenario labels and gaps), summary (per-scenario estimates), mcse (within-scenario Monte Carlo SE), trials (individual trials), and aggregate (mean and SD across scenarios grouped separately by method, MTD category and gap combination).

Examples

```
z <- simulate_random_scenarios(mtd=2, upper_gap=.15,
  nscenarios=2, nsim=5, method="prediction", seed=42)
z$settings
z$aggregate
```

`simulate_scenarios` *Run a panel of dose-toxicity scenarios*

Description

Results include scenario-level estimates and mean/SD summaries across random scenarios within each method and MTD category. Use explicit MTD labels for all-too-toxic and below-range cases.

Usage

```
simulate_scenarios(
  scenarios,
  mtd = NULL,
  method = c("prediction", "tite", "combined"),
  config = bold_config(),
  nsim = 1000,
  seed = 1
)
```

Arguments

`scenarios`	Numeric matrix with one nondecreasing risk vector per row.
`mtd`	Optional integer vector giving each scenario's optimal dose.
`method`, `config`, `nsim`, `seed`	As in simulate_bold(). Seeds increment by scenario index; methods within a scenario share the same seed.

Value

List with summary, mcse and trials data frames containing scenario, and aggregate containing means and between-scenario standard deviations.

Examples

```
x <- rbind(c(.1,.25,.45,.6),c(.05,.1,.25,.45))
simulate_scenarios(x, mtd=c(2,3), method="prediction", nsim=5)$summary
```

`summarise_scenarios` *Summarise operating characteristics across random scenarios*

Description

Compute the equally weighted mean and sample standard deviation of scenario-level estimates, each obtained from its simulated trial replicates.

Usage

```
summarise_scenarios(x, group_by = c("method", "mtd"))
```

Arguments

`x`	Scenario-level summary data, or a scenario simulation result.
`group_by`	Reporting-group columns, including method. Additional columns, such as delta, can be attached to x before summarising.

Details

The SD measures between-scenario dispersion, not a standard error. For 50 scenarios with 100 replicates each, the SD is calculated from the 50 scenario-level estimates, not from the 5,000 individual trials. It includes both scenario heterogeneity and finite-replicate simulation noise. Group by MTD category, upper toxicity gap, and design settings where appropriate. A group with only one scenario has undefined SD, reported as NA. Above-MTD percentages are first patient-weighted within each scenario, then equally weighted across scenarios. Missing duration for complete-follow-up methods is retained as NA.

Value

A data frame of point-estimate means and matching *_sd columns, with n_scenarios and the minimum/maximum replicates per scenario.

Calculation

For J scenario-level estimates Q_j, the reported values are

$$\overline{Q} = J^{-1}\sum_{j=1}^{J} Q_j, \qquad SD(Q) = \sqrt{\frac{\sum_{j=1}^{J}(Q_j - \overline{Q})^2}{J-1}}.$$

Examples

```
panel <- simulate_scenarios(rbind(c(.1,.25,.45), c(.15,.25,.4)),
  mtd=c(2,2), method="prediction", nsim=5)
panel$aggregate
```

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